Problem Set 5
Regression Analysis with Time Series Data
Econometrics

1. Decide if you agree or disagree with each of the following statements and give a brief explanation of your decision:

   (a) Like cross-sectional observations, we can assume that most time series observations are independently distributed.

   (b) The OLS estimator in a time series regression is unbiased under the first three Gauss-Markov assumptions.

   (c) A trending variable cannot be used as the dependent variable in multiple regression analysis.

   (d) Seasonality is not an issue when using annual time series observations.

   Solution:

   (a) Disagree. Most time series processes are correlated over time, and many of them strongly correlated. This means they cannot be independent across observations, which simply represent different time periods. Even series that do appear to be roughly uncorrelated - such as stock returns - do not appear to be independently distributed, as you will see in Chapter 12 under dynamic forms of heteroskedasticity.

   (b) Agree. We do not need the homoskedasticity and no serial correlation assumptions.

   (c) Disagree. Trending variables are used all the time as dependent variables in a regression model. We do need to be careful in interpreting the results because we may simply find a spurious association between y_t and trending explanatory variables. Including a trend in the regression is a good idea with trending dependent or independent variables. The usual R-squared can be misleading when the dependent variable is trending.

   (d) Agree. With annual data, each time period represents a year and is not associated with any season.

2. Let $gGDP_t$ denote the annual percentage change in gross domestic product and let int_t denote a shortterm interest rate. Suppose that $gGDP_t$ is related to interest rates by

$$gGDP_t = \alpha_0 + \delta_0 int_t + \delta_1 iit_{t-1} + u_t$$

   where u_t is uncorrelated with int$_t$, int t_{t-1}, and all other past values of interest rates. Suppose that the Federal Reserve follows the policy rule:

$$int_t = \gamma_0 + \gamma_1 \left(gGDP_{t-1} - 3\right) + v_t$$

where $\gamma_1 > 0$. (When last year's GDP growth is above 3%, the Fed increases interest rates to prevent an "overheated" economy.) If v_t is uncorrelated with all past values of int_t and u_t, argue that int $_t$ must be correlated with u_{t-1}. (Hint: Lag the first equation for one time period and substitute for $gGDP_{t-1}$ in the second equation.) Which Gauss-Markov assumption does this violate?

Solution:

(a) Start by lagging the first equation by one period:

$$gGDP_{t-1} = \alpha_0 + \delta_0 \cdot int_{t-1} + \delta_1 \cdot int_{t-2} + u_{t-1}.$$

(b) Next, substitute the expression for $gGDP_{t-1}$ into the second equation (the policy rule):

$$int_t = \gamma_0 + \gamma_1 \left[(\alpha_0 + \delta_0 \cdot int_{t-1} + \delta_1 \cdot int_{t-2} + u_{t-1}) - 3 \right] + v_t.$$

Simplifying, we get:

$$int_t = \gamma_0 + \gamma_1 (\alpha_0 - 3) + \gamma_1 \cdot \delta_0 \cdot int_{t-1} + \gamma_1 \cdot \delta_1 \cdot int_{t-2} + \gamma_1 \cdot u_{t-1} + v_t.$$

(c) Notice that int_t now depends directly on u_{t-1} through the term $\gamma_1 \cdot u_{t-1}$. Therefore, int_t is correlated with u_{t-1}.

(d) The implication is that the error term u_t in the original model is correlated with int_t. Specifically, since int_t depends on u_{t-1}, and u_{t-1} is part of the error term from the previous period, there is a correlation between the error term u_t and one of the explanatory variables int_t.

(e) **Violation of Gauss-Markov Assumptions**: The Gauss-Markov assumption violated here is the assumption of no serial correlation. In other words, the assumption that the error term u_t is uncorrelated with itself across different time periods is violated because int_t is correlated with u_{t-1}. When this assumption is violated, the OLS estimators are no longer BLUE (Best Linear Unbiased Estimators).

3. Suppose you have quarterly data on new housing starts, interest rates, and real per capita income. Specify a model for housing starts that accounts for possible trends and seasonality in the variables.

Solution:

Let H_t represent the number of new housing starts in quarter t, r_t represent the interest rate in quarter t, and Y_t represent the real per capita income in quarter t.

The model can be specified as:

$$H_t = \beta_0 + \beta_1 r_t + \beta_2 Y_t + \beta_3 t + \sum_{j=1}^{3} \gamma_j Q_{jt} + u_t,$$

where:

- β_0 is the intercept.
- r_t is the interest rate, with coefficient β_1.
- Y_t is the real per capita income, with coefficient β_2.

- t is a time trend, included to account for any long-term trend in housing starts, with coefficient β_3.
- Q_{jt} are dummy variables for quarters 1 through 3 (where $Q_{1t} = 1$ if the observation is in quarter 1, and 0 otherwise, similarly for quarters 2 and 3). Quarter 4 is treated as the base category, so its effect is captured in the intercept β_0. The coefficients γ_1, γ_2, and γ_3 capture seasonal effects.
- u_t is the error term.

Explanation:

- The inclusion of the time trend t allows the model to capture any systematic growth or decline in housing starts over time that is not explained by interest rates or real per capita income. For example, if the housing market has been growing steadily over time, the trend variable will capture this growth.
- The dummy variables Q_{1t}, Q_{2t}, and Q_{3t} account for seasonality in housing starts. In many cases, housing starts are higher in certain seasons (e.g., spring and summer) than in others (e.g., winter). These quarterly dummies allow the model to adjust for such predictable seasonal patterns.
- The interest rate r_t is expected to have a negative effect on housing starts, as higher interest rates increase the cost of borrowing, thereby reducing the demand for new homes.
- Real per capita income Y_t is expected to have a positive effect on housing starts, as higher income levels generally lead to greater demand for housing.

Model Interpretation:

- β_1: Measures the impact of a one-unit increase in the interest rate on housing starts, holding other factors constant.
- β_2: Measures the impact of a one-unit increase in real per capita income on housing starts, holding other factors constant.
- β_3: Captures the long-term trend in housing starts over time.
- $\gamma_1, \gamma_2, \gamma_3$: Measure the seasonal effects in quarters 1, 2, and 3 relative to quarter 4.

4. The index of industrial production (IP) is a monthly time series that measures the quantity of industrial commodities produced in a given month. This problem uses data on this index for the United States. Let $Y_t = 1200 \times \ln(IP_t/IP_{t-1})$.

   (a) The forecaster states that Y_t shows the monthly percentage change in IP, measured in percentage points per year. Is this correct? Why?

   (b) Suppose a forecaster estimates an auto-regressive model with 4 lags (AR(4)) for Y_t:

$$\hat{Y}_t = \underset{(0.062)}{1.377} + \underset{(0.078)}{0.318}Y_{t-1} + \underset{(0.055)}{0.123}Y_{t-2} + \underset{(0.068)}{0.068}Y_{r-3} + \underset{(0.056)}{0.001}Y_{t-t}.$$

   Use this AR(4) to forecast the value of Y_t in January 2001 using the following values of industrial production for July 2000 through December 2000):

Date	2000-7	2000-8	2000-9	2000-10	2000-11	2000-12
IP	147.595	148.650	148.973	148.660	148.206	147.300

(c) Worried about potential seasonal fluctuations in production. the forecaster adds Y_{t-12} to the model. The estimated coefficient is Y_{t-12} is -0.054 with a standard error of 0.053. Is this coefficient statistically significant?

Solution:

(a) The term $\ln(IP_t) - \ln(IP_{t-1})$ represents the monthly percentage change in IP, but expressed in logarithmic terms. The factor 1200 is used to annualize this percentage change (since there are 12 months in a year), converting it into percentage points per year.

Thus, the forecaster's statement is correct: Y_t represents the monthly percentage change in IP, annualized to reflect percentage points per year.

(b) First, we calculate Y_t for each month from August 2000 to December 2000 using:

$$Y_t = 1200 \times \ln\left(\frac{IP_t}{IP_{t-1}}\right).$$

The calculations are as follows:

$$Y_{\text{2000-8}} = 1200 \times \ln\left(\frac{148.650}{147.595}\right) \approx 8.544,$$

$$Y_{\text{2000-9}} = 1200 \times \ln\left(\frac{148.973}{148.650}\right) \approx 2.597,$$

$$Y_{\text{2000-10}} = 1200 \times \ln\left(\frac{148.660}{148.973}\right) \approx -2.527,$$

$$Y_{\text{2000-11}} = 1200 \times \ln\left(\frac{148.206}{148.660}\right) \approx -3.665,$$

$$Y_{\text{2000-12}} = 1200 \times \ln\left(\frac{147.300}{148.206}\right) \approx -7.355.$$

Next, we use the AR(4) model to forecast Y_t for January 2001:

$$\hat{Y}_{\text{2001-01}} = 1.377 + 0.318 \times (-7.355) + 0.123 \times (-3.665) + 0.068 \times (-2.527) + 0.001 \times 2.597.$$

Calculating step-by-step:

$$\hat{Y}_{\text{2001-01}} = 1.377 - 2.339 + (-0.451) + (-0.172) + 0.003 \approx -1.582.$$

Therefore, the forecasted value for January 2001 is $\hat{Y}_{\text{2001-01}} \approx -1.582$.

(c) The model with the additional lag Y_{t-12} is given by:

(d) The model with the additional lag Y_{t-12} is given by:

$$\hat{Y}_t = 1.377 + 0.318Y_{t-1} + 0.123Y_{t-2} + 0.068Y_{t-3} + 0.001Y_{t-4} - 0.054Y_{t-12}.$$

The standard error of the coefficient on Y_{t-12} is 0.053.

To determine if this coefficient is statistically significant, we perform a t-test:

$$t = \frac{-0.054}{0.053} \approx -1.02.$$

At a 5% significance level, the critical value for a two-tailed test with a large sample size is approximately 1.96. Since $|t| < 1.96$, the coefficient on Y_{t-12} is not statistically significant, meaning it does not have a strong enough impact to warrant its inclusion in the model.

Computer Exercises

1. Use the data in VOLAT for this exercise. The variable $rsp500$ is the monthly return on the Standard & Poor's 500 stock market index, at an annual rate. (This includes price changes as well as dividends.) The variable $i3$ is the return on three-month T-bills, and pcip is the percentage change in industrial production; these are also at an annual rate.

   (a) Consider the equation
   $$rsp500_t = \beta_0 + \beta_1 pcip_t + \beta_2 i3_t + u_t$$
   What signs do you think β_1 and β_2 should have?

   (b) Estimate the previous equation by OLS, reporting the results in standard form. Interpret the signs and magnitudes of the coefficients.

   (c) Which of the variables is statistically significant?

   (d) Does your finding from part (iii) imply that the return on the S&P 500 is predictable? Explain.

Solution:

(a) - For β_1 (coefficient of pcip): We expect it to be positive because higher industrial production growth (pcip) signals better economic conditions, which should lead to higher returns on the S&P 500.

- For β_2 (coefficient of i3): We expect it to be negative because an increase in short-term interest rates (i3) makes risk-free assets more attractive, reducing the demand for stocks and potentially leading to lower returns.

(b)
```
##############################################3
data(volat, package='wooldridge')

# Part (ii): Estimate the equation by OLS and report the results
model <- lm(rsp500 ~ pcip + i3, data = volat)
summary(model)

coefficients <- coef(summary(model))
r_squared <- summary(model)$r.squared
```

```r
# Print key results
cat("Estimated Equation:\n")
cat("rsp500 = ", round(coefficients[1, 1], 4), " + ", round(coefficients[2,
    1], 4), " * pcip + ", round(coefficients[3, 1], 4), " * i3\n\n")
cat("P-values:\n")
cat("pcip: ", round(coefficients[2, 4], 4), "\n")
cat("i3: ", round(coefficients[3, 4], 4), "\n\n")
cat("R-squared: ", round(r_squared, 4), "\n")

# Interpretation of predictability
if (coefficients[2, 4] < 0.05 | coefficients[3, 4] < 0.05) {
  cat("At least one of the predictors is statistically significant, 
     suggesting that the return on the S&P 500 may be predictable.\n")
} else {
  cat("Neither predictor is statistically significant, so there is no 
     strong evidence of predictability.\n")
}
```

2. The file TRAFFIC2 contains 108 monthly observations on automobile accidents, traffic laws, and some other variables for California from January 1981 through December 1989. Use this data set to answer the following questions.

   (a) During what month and year did California's seat belt law take effect? When did the highway speed limit increase to 65 miles per hour?

   (b) Regress the variable log(totacc) on a linear time trend and 11 monthly dummy variables, using January as the base month. Interpret the coefficient estimate on the time trend. Would you say there is seasonality in total accidents?

   (c) Add to the regression from part (b) the variables wkends, unem, spdlaw, and beltlaw. Discuss the coefficient on the unemployment variable. Does its sign and magnitude make sense to you?

   (d) In the regression from part (c), interpret the coefficients on spdlaw and beltlaw. Are the estimated effects what you expected? Explain.

   (e) The variable prcfat is the percentage of accidents resulting in at least one fatality. Note that this variable is a percentage, not a proportion. What is the average of prcfat over this period? Does the magnitude seem about right?

   (f) Run the regression in part (iii) but use prcfat as the dependent variable in place of log(totacc). Discuss the estimated effects and significance of the speed and seat belt law variables.

Solution:

```r
#####################################################
data(traffic2, package = "wooldridge")

# Load necessary packages
library(dplyr)
library(lubridate)

# Load and inspect the dataset
head(traffic2)
```

```r
# Part (i): During what month and year did California's seat belt law take
    effect? When did the highway speed limit increase to 65 miles per hour?

# Seat belt law: The variable 'beltlaw' indicates when the seat belt law took
    effect (beltlaw = 1).
# Speed limit increase: The variable 'spdlaw' indicates when the speed limit
    increased to 65 mph (spdlaw = 1).

# Extract the month and year when these laws took effect
seat_belt_law_date <- traffic2 %>% filter(beltlaw == 1) %>% select(year) %>%
    distinct()
speed_limit_date <- traffic2 %>% filter(spdlaw == 1) %>% select(year) %>%
    distinct()

cat("Seat␣belt␣law␣took␣effect␣in:", seat_belt_law_date$year, "\n")
cat("Highway␣speed␣limit␣increased␣to␣65␣mph␣in:", speed_limit_date$year, "\n")

# Part (ii): Regress log(totacc) on the linear time trend (t) and monthly dummy
     variables (feb to dec)
model_part_ii <- lm(log(totacc) ~ t + feb + mar + apr + may + jun + jul + aug +
     sep + oct + nov + dec, data = traffic2)
summary(model_part_ii)

# Part (iii): Regress log(totacc) on the time trend (t), monthly dummy
    variables (feb to dec), and additional covariates (wkends, unem, spdlaw,
    beltlaw)
model_part_iii <- lm(log(totacc) ~ t + feb + mar + apr + may + jun + jul + aug
    + sep + oct + nov + dec + wkends + unem + spdlaw + beltlaw, data = traffic2)
summary(model_part_iii)

# Part (iv): Interpret the coefficients on spdlaw and beltlaw in the regression
     from part (iii).

# The coefficient on spdlaw shows the effect of increasing the speed limit to
    65 mph on total accidents.
# The coefficient on beltlaw shows the effect of introducing the seat belt law
    on total accidents.
# Discuss whether the signs of these coefficients match expectations (e.g., a
    positive sign for spdlaw indicating more accidents due to higher speed
    limits).

# Part (v): Calculate the average of prcfat and discuss whether the magnitude
    seems reasonable.

avg_prcfat <- mean(traffic2$prcfat, na.rm = TRUE)
cat("Average␣prcfat:", round(avg_prcfat, 2), "%\n")

# The average percentage of accidents resulting in at least one fatality is
    about X%. Discuss whether this percentage seems reasonable based on the
    context.

# Part (vi): Regress prcfat on the time trend (t), monthly dummy variables (feb
     to dec), and additional covariates (wkends, unem, spdlaw, beltlaw)
model_prcfat <- lm(prcfat ~ t + feb + mar + apr + may + jun + jul + aug + sep +
     oct + nov + dec + wkends + unem + spdlaw + beltlaw, data = traffic2)
summary(model_prcfat)
```

```
# Discuss the estimated effects of spdlaw and beltlaw on prcfat.
# If spdlaw has a significant positive coefficient, it suggests that increasing
    the speed limit leads to a higher percentage of fatal accidents.
# If beltlaw has a significant negative coefficient, it suggests that the seat
   belt law reduces the percentage of fatal accidents.
```